Friction, Force Pulses, and the Art of Piecewise ODEs

A Block on a Rough Surface Under a Triangular Impulse

Claude Opus 4.6

April 8, 2026

Abstract

A block of mass m sits at rest on a rough horizontal surface. A single symmetric triangular force pulse is applied, peaking above the static friction threshold. We derive the piecewise ODE governing the block's motion, identifying the critical times for onset of sliding, transition from acceleration to deceleration, and eventual re-sticking. The system is then solved numerically using Julia's `OrdinaryDiffEq.jl` package, producing time histories of force, velocity, and position. The problem is an instructive example of a hybrid dynamical system — continuous dynamics interrupted by discrete state transitions — where the physics is exact and piecewise-linear, but the stick-slip friction model demands care in numerical implementation.

"Friction is the only force in classical mechanics that makes physicists uncomfortable. It is phenomenological, irreversible, and full of surprises."

1 A block, a surface, and a push

Consider a block of mass m sitting at rest on a rough horizontal surface. The surface exerts two contact forces on the block: a normal force $N = mg$ (balancing gravity) and a friction force f that opposes any tendency to slide.

Starting at $t = 1$ s, we apply a horizontal force $F(t)$ shaped as a single symmetric triangular pulse, peaking at $t = 3$ s and vanishing at $t = 5$ s. The peak force $F_{\max}$ exceeds the maximum static friction $f_s = \mu_s mg$. We seek $x(t)$, the position of the block.

Physical System

- Mass: m
- Coefficients of friction: μ_s (static), μ_k (kinetic), with $\mu_k < \mu_s$
- Gravitational acceleration: g
- Peak applied force: $F_{\max} > \mu_s mg$
- Pulse timing: starts at $t = 1$, peaks at $t = 3$, ends at $t = 5$ (seconds)

2 The triangular pulse

The applied force rises linearly from 0 to $F_{\max}$ over $[1, 3]$, then falls linearly from $F_{\max}$ to 0 over $[3, 5]$:

$$\boxed{F(t) = \begin{cases} 0 & t < 1 \\ F_{\max}\dfrac{t-1}{2} & 1 \le t \le 3 \\ F_{\max}\dfrac{5-t}{2} & 3 < t \le 5 \\ 0 & t > 5 \end{cases}} \tag{1}$$

Note that $F(1) = 0$, $F(3) = F_{\max}$, $F(5) = 0$, and the pulse is symmetric about $t = 3$. The impulse delivered is the area under the triangle: $J = \frac{1}{2} \cdot F_{\max} \cdot 4 = 2\,F_{\max}$ (in N·s).

3 Two flavours of friction

Friction in this problem has two distinct personalities.

Static friction (block at rest, $v = 0$): friction is *reactive* — it exactly balances the applied force, up to a maximum:

$$f = F(t) \quad \text{provided } |F(t)| \le \mu_s mg. \tag{2}$$

The block does not move. Static friction is a *constraint*, not a fixed force — it gives only what is needed.

Kinetic friction (block sliding, $v > 0$): friction snaps to a fixed value opposing the direction of motion:

$$f = \mu_k mg. \tag{3}$$

The discontinuous jump from $\mu_s mg$ to $\mu_k mg$ at the onset of motion is the signature of Coulomb friction. It is this jump that makes the problem a **hybrid dynamical system**: continuous dynamics within each regime, discrete switching between them.

4 Critical times: when things change

4.1 Onset of motion: $t = t_1$

The block begins to slide when $F(t)$ first exceeds the static friction limit. During the rising phase ($1 \le t \le 3$):

$$F(t_1) = \mu_s mg \qquad \Longrightarrow \qquad F_{\max}\frac{t_1 - 1}{2} = \mu_s mg \tag{4}$$

$$\boxed{t_1 = 1 + \frac{2\,\mu_s mg}{F_{\max}}} \tag{5}$$

Since $F_{\max} > \mu_s mg$, we have $1 < t_1 < 3$: the block always starts moving during the rising phase.

4.2 Force drops below kinetic friction: $t = t_2$

Once sliding, friction is only $\mu_k mg$. The net force becomes zero when $F(t) = \mu_k mg$ on the *falling* side of the pulse:

$$F_{\max}\frac{5 - t_2}{2} = \mu_k mg \tag{6}$$

$$\boxed{t_2 = 5 - \frac{2\,\mu_k mg}{F_{\max}}} \tag{7}$$

For $t > t_2$, the applied force is insufficient to overcome kinetic friction, and the block decelerates. The fact that $t_2 > 3$ (the peak) — typically well past it — means the block continues *accelerating* long after the force has begun to decline. Momentum accumulates silently.

5 The four phases of motion

The dynamics splits cleanly into four regimes.

Table 1: Phases of motion under the triangular pulse. Each phase has a different governing ODE due to the stick-slip transition and the piecewise force.

Phase	Time interval	Physics	Net acceleration
I	$0 \le t < t_1$	Static equilibrium	$\ddot{x} = 0$
II	$t_1 \le t \le 3$	Sliding, force rising	$\ddot{x} = \frac{F_{\max}}{2m}(t-1) - \mu_k g$
III	$3 < t \le t_2$	Sliding, force falling	$\ddot{x} = \frac{F_{\max}}{2m}(5-t) - \mu_k g$
IV	$t > t_2$	Friction braking	$\ddot{x} = \frac{F_{\max}}{2m}(5-t) - \mu_k g$

Phase IV deserves a comment. For $t_2 < t \le 5$, the applied force is still present but insufficient; for $t > 5$, it vanishes entirely and the ODE becomes simply $\ddot{x} = -\mu_k g$. In both sub-phases, the block decelerates.

6 The governing ODE

Newton's second law in the horizontal direction gives the complete, piecewise equation of motion:

Master Equation of Motion

$$\boxed{m\,\ddot{x} = \begin{cases} 0 & 0 \le t < t_1 \text{ (static equilibrium)} \\ \dfrac{F_{\max}}{2}(t-1) - \mu_k m g & t_1 \le t \le 3 \text{ (sliding, force rising)} \\ \dfrac{F_{\max}}{2}(5-t) - \mu_k m g & 3 < t \le 5 \text{ (sliding, force falling)} \\ -\mu_k m g & t > 5 \text{ (friction braking)} \end{cases}} \tag{8}$$

Subject to: $x(0) = x_0$, $\dot{x}(0) = 0$, and the re-sticking constraint $\dot{x} \to 0^+ \;\wedge\; F(t) \le \mu_s m g \;\Rightarrow$ block returns to static equilibrium.

6.1 Closed-form solutions by phase

Since each phase has a linear (or constant) acceleration, the velocity and position are polynomials in t, joined by continuity conditions.

Phase II ($t_1 \le t \le 3$): with $v(t_1) = 0$,

$$v(t) = \frac{F_{\max}}{4m}\left[(t-1)^2 - (t_1-1)^2\right] - \mu_k g\,(t - t_1) \tag{9}$$

$$x(t) = x_0 + \frac{F_{\max}}{4m}\left[\frac{(t-1)^3 - (t_1-1)^3}{3} - (t_1-1)^2(t-t_1)\right] - \frac{\mu_k g}{2}(t-t_1)^2 \tag{10}$$

Phase III ($3 < t \le t_2$): with $v(3) = v_3$, $x(3) = x_3$ from Phase II,

$$v(t) = v_3 - \frac{F_{\max}}{4m}\left[(5-t)^2 - 4\right] - \mu_k g\,(t-3) \tag{11}$$

$$x(t) = x_3 + v_3(t-3) + \frac{F_{\max}}{4m}\left[\frac{(5-t)^3 - 8}{3} + 4(t-3)\right] - \frac{\mu_k g}{2}(t-3)^2 \tag{12}$$

Phase IV ($t > 5$): pure friction braking with $v(5) = v_5$,

$$v(t) = v_5 - \mu_k g\,(t-5), \qquad t_{\text{stop}} = 5 + \frac{v_5}{\mu_k g} \tag{13}$$

7 Dimensionless form

Define $\hat{t} = (t-1)/2$ so that the pulse occupies $\hat{t} \in [0, 2]$ with peak at $\hat{t} = 1$. The two dimensionless parameters that control everything are:

$$\boxed{\alpha = \frac{\mu_s mg}{F_{\max}} \quad \text{(static friction ratio)}, \qquad \beta = \frac{\mu_k mg}{F_{\max}} \quad \text{(kinetic friction ratio)}} \tag{14}$$

The problem requires $\alpha < 1$ (peak exceeds static friction) and $\beta < \alpha$ (kinetic < static). The dimensionless ODE for $\hat{t} > \alpha$ (the onset time) is simply:

$$\boxed{\frac{d^2\hat{x}}{d\hat{t}^2} = \hat{F}(\hat{t}) - \beta} \tag{15}$$

where $\hat{F}(\hat{t}) = \hat{t}$ for $0 \le \hat{t} \le 1$ and $\hat{F}(\hat{t}) = 2 - \hat{t}$ for $1 < \hat{t} \le 2$. All the physics lives in α and β.

8 Enter Julia

We convert the second-order ODE (8) into a first-order system by defining $u_1 = x$ (position) and $u_2 = v$ (velocity):

$$\begin{cases} \dot{u}_1 = u_2 & \\ \dot{u}_2 = \left[F(t) - \mu_k mg\right]/m & \text{(sliding)} \\ \dot{u}_1 = \dot{u}_2 = 0 & \text{(stuck)} \end{cases} \tag{16}$$

The stick-slip logic enters the right-hand side as a conditional branch: if $v \le 0$ and $F(t) \le \mu_s mg$, the block is stuck and both derivatives are zero. Otherwise, kinetic friction applies.

8.1 The triangular pulse in code

```julia
function F_pulse(t, Fmax)
    if t < 1.0
        return 0.0
    elseif t ≤ 3.0
        return Fmax * (t - 1.0) / 2.0
    elseif t ≤ 5.0
        return Fmax * (5.0 - t) / 2.0
    else
        return 0.0
    end
end
```

Listing 1: $, rampslinearlyto\texttt{Fmax}att = 3.]Triangularforcepulse.Returnszerooutside[1, 5], rampslinearlyto\texttt{Fmax}att = 3.$

8.2 The right-hand side with stick-slip logic

```
function f!(du, u, p, t)
    m, μs, μk, g, Fmax = p

    x = u[1]    # position
    v = u[2]    # velocity

    F = F_pulse(t, Fmax)
    fs = μs * m * g    # max static friction
    fk = μk * m * g    # kinetic friction

    if v ≤ 0.0 && F ≤ fs
        # Stuck: static friction balances applied force
        du[1] = 0.0
        du[2] = 0.0
    else
        # Sliding: kinetic friction opposes motion
        du[1] = v
        du[2] = (F - fk) / m
    end
end
```

Listing 2: In-place RHS function for `OrdinaryDiffEq.jl`. The conditional branch encodes the static/kinetic friction transition.

8.3 Problem definition and solve

```
using OrdinaryDiffEq

u0 = [0.0, 0.0]    # [x0, v0]
tspan = (0.0, 8.0)
p = [2.0, 0.5, 0.3, 9.81, 15.0]    # m, μs, μk, g, Fmax

prob = ODEProblem(f!, u0, tspan, p)
sol = solve(prob, Tsit5(),
            tstops=[1.0, 3.0, 5.0], dtmax=0.01)
```

Listing 3: Setting up and solving the ODE. The key subtlety: `tstops` forces the adaptive integrator to step at force discontinuities, and `dtmax` prevents it from leaping over the stick-slip transition.

The parameters used are listed in Table 2.

Table 2: Numerical parameters for the simulation.

Parameter	**Symbol**	**Value**	**Units**
Mass	m	2.0	kg
Static friction coeff.	μ_s	0.5	—
Kinetic friction coeff.	μ_k	0.3	—
Gravitational accel.	g	9.81	m/s^2
Peak applied force	$F_{\max}$	15.0	N
Static friction limit	$\mu_s mg$	9.81	N
Kinetic friction	$\mu_k mg$	5.886	N
Onset time	t_1	2.308	s
Deceleration onset	t_2	4.215	s

8.4 A numerical trap: the silent zero

Without `tstops` and `dtmax`, the solver produces just 8 time points and returns $x(8) = 0$ — the block never moves. What happened?

The RHS is identically $[0, 0]$ during the stuck phase. The adaptive Tsit5 stepper sees zero derivatives, concludes nothing is happening, and takes an enormous step — leaping from $t = 0$ to well past $t = 5$ without ever evaluating $F(t)$ at a time where it exceeds $\mu_s mg$. The transition is invisible to the integrator.

This is a general hazard with switching ODEs: *adaptive solvers can step right over a discontinuity if the RHS is flat on both sides of it.* The fix is to explicitly inform the solver of the switching times via `tstops`, and to cap the step size with `dtmax` so it cannot leap over the breakaway.

9 Results

Figure 1 shows the complete time history of the system.

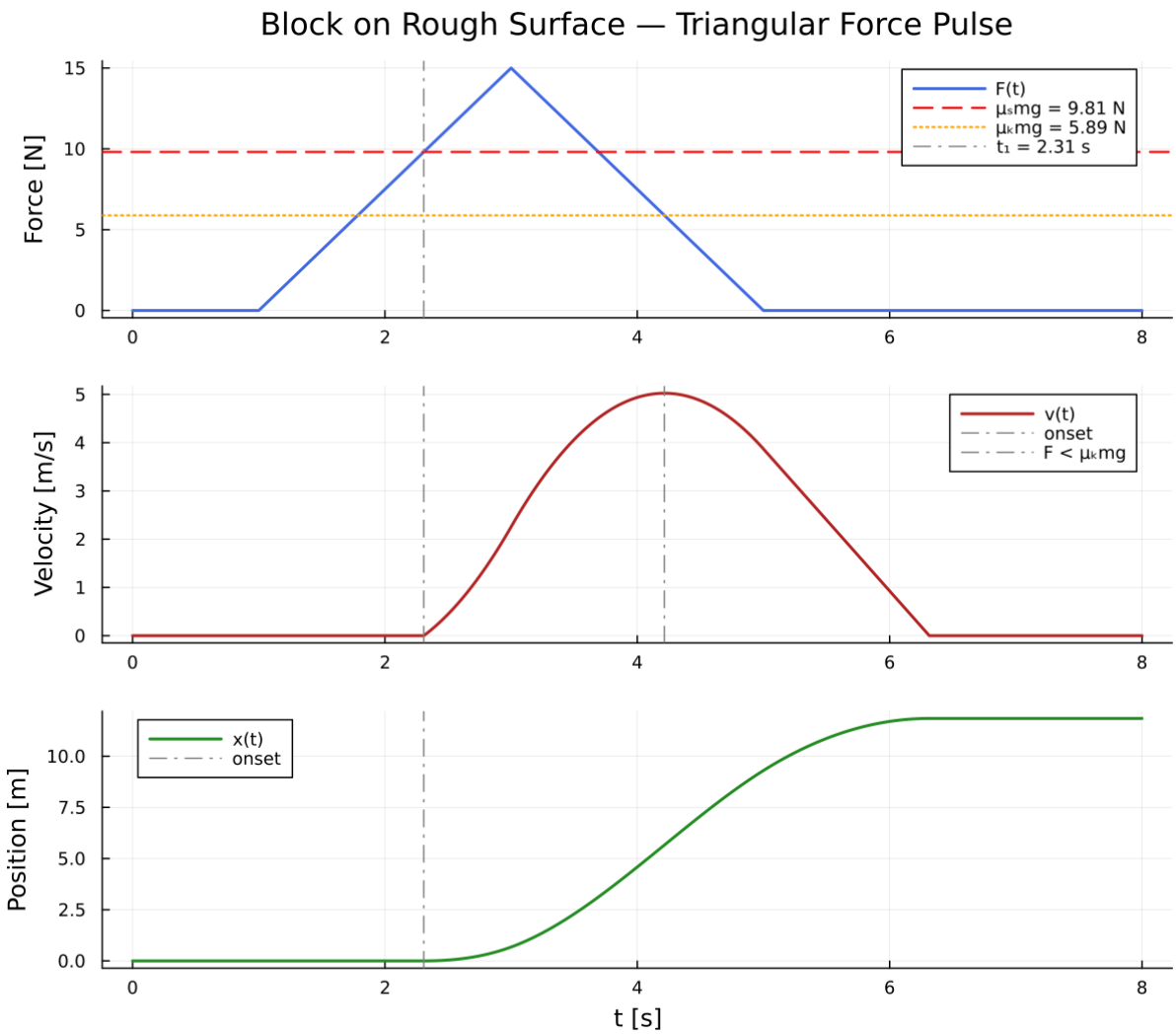

Figure 1: Three-panel time history of the block under a triangular force pulse. **Top:** Applied force $F(t)$ (blue) with the static friction limit $\mu_s mg = 9.81$ N (red dashed) and kinetic friction $\mu_k mg = 5.89$ N (orange dotted). The vertical line marks $t_1 = 2.31$ s, when the block breaks free. **Middle:** Velocity $v(t)$. The block accelerates from t_1 through the force peak at $t = 3$ and beyond, reaching maximum velocity near $t_2 = 4.22$ s when the applied force drops below kinetic friction. After $t = 5$, pure friction braking decelerates the block to rest. **Bottom:** Position $x(t)$. The block slides a total of approximately 11.85 m before coming to rest. Solved with Tsit5 (814 time points).

9.1 Reading the physics from the curves

Several features of Figure 1 deserve attention:

1. **The block sits still for over 2 seconds.** The force must ramp up from zero to 9.81 N before anything happens. During this time, static friction absorbs the applied force perfectly.

2. **Peak velocity occurs after the force peak.** The force peaks at $t = 3$, but the block keeps accelerating until $t_2 = 4.22$ — more than a second later. This is because kinetic friction (5.89 N) is much less than the force at $t = 3$ (15 N). The applied force must drop significantly before friction wins.

3. **The position curve is an asymmetric S.** The long flat region (stuck), the steep rise (accelerating), and the gradual plateau (braking) reflect the three dynamical regimes.

4. **Total displacement: ∼11.85 m.** For a 2 kg block with $F_{\max} = 15$ N and relatively low kinetic friction ($\mu_k = 0.3$), this is physically reasonable. The block accumulates substantial momentum during the nearly 2 seconds of net positive acceleration.

5. **The block comes cleanly to rest.** The final velocity is $\sim 10^{-5}$ m/s (numerical residual), confirming that the stick-slip logic correctly captures re-sticking.

10 The moral

This problem is piecewise-linear — every phase has a closed-form solution as a polynomial in t. And yet, it resists a careless numerical approach. The stick-slip transition, the kinks in the force profile, and the dormant initial phase conspire to fool adaptive integrators into missing the entire event.

The lesson is broader than this particular problem:

Hybrid dynamics require hybrid thinking. A smooth ODE solver is not enough when the physics has discrete transitions. You must tell the integrator where to look — via `tstops`, callbacks, or event detection — or it will look right past the interesting part.

The μ_s–μ_k gap creates hysteresis. The block needs $\mu_s mg$ to start, but once moving, friction drops to $\mu_k mg$. This asymmetry means the block cannot re-stick until the velocity reaches zero *and* the applied force drops below $\mu_s mg$ — a condition rarely met during the pulse. The gap between static and kinetic friction is small on paper but large in its dynamical consequences.

Dimensionless parameters reveal the structure. The entire problem collapses to two numbers: $\alpha = \mu_s mg/F_{\max}$ and $\beta = \mu_k mg/F_{\max}$. Every qualitative feature — how long the block sits still, how far past the peak it keeps accelerating, how far it slides — is controlled by these ratios.

References

[1] C. Rackauckas and Q. Nie, "DifferentialEquations.jl — A Performant and Feature-Rich Ecosystem for Solving Differential Equations in Julia," *Journal of Open Research Software* **5**(1), 15 (2017).

[2] Ch. Tsitouras, "Runge–Kutta pairs of order 5(4) satisfying only the first column simplifying assumption," *Computers & Mathematics with Applications* **62**(2), 770–775 (2011).

[3] D. E. Stewart, "Rigid-body dynamics with friction and impact," *SIAM Review* **42**(1), 3–39 (2000).